

Vector-Valued Functions

ANSWER KEY



Position, velocity, and acceleration vectors

- 1 A particle's position is given by $\vec{r}(t) = \langle t^2, t^3 \rangle$. Find the velocity vector $\vec{v}(t)$ and acceleration vector $\vec{a}(t)$.
 $\vec{v}(t) = \langle 2t, 3t^2 \rangle$. $\vec{a}(t) = \langle 2, 6t \rangle$.
- 2 For the same particle, find the speed at $t = 2$.
 $\vec{v}(2) = \langle 4, 12 \rangle$. Speed $= |\vec{v}(2)| = \sqrt{4^2 + 12^2} = \sqrt{16 + 144} = \sqrt{160} = 4\sqrt{10}$.

Integrating vector-valued functions

- 3 Given $\vec{v}(t) = \langle 3t^2, 2t \rangle$ and initial position $\vec{r}(0) = \langle 1, 4 \rangle$, find $\vec{r}(t)$.
 $\vec{r}(t) = \langle t^3 + C_1, t^2 + C_2 \rangle$. Using $\vec{r}(0) = \langle 1, 4 \rangle$: $C_1 = 1$, $C_2 = 4$. $\vec{r}(t) = \langle t^3 + 1, t^2 + 4 \rangle$.

Total distance traveled

- 4 A particle has velocity $\vec{v}(t) = \langle \cos t, \sin t \rangle$ for $t \in [0, \pi]$. Find the total distance traveled.
Speed $= |\vec{v}(t)| = \sqrt{\cos^2 t + \sin^2 t} = 1$ (constant). Distance $= \int_0^\pi 1 \, dt = \pi$.

Vectors and parametric motion combined

- 5 A particle moves along $x(t) = e^t$, $y(t) = e^{-t}$. Find the velocity vector and speed at $t = 0$.
 $\vec{v}(t) = \langle e^t, -e^{-t} \rangle$. At $t = 0$: $\vec{v}(0) = \langle 1, -1 \rangle$. Speed $= \sqrt{1^2 + (-1)^2} = \sqrt{2}$.
- 6 For the particle above, find the magnitude of the acceleration vector at $t = 0$.
 $\vec{a}(t) = \langle e^t, e^{-t} \rangle$. At $t = 0$: $\vec{a}(0) = \langle 1, 1 \rangle$. $|\vec{a}(0)| = \sqrt{1^2 + 1^2} = \sqrt{2}$.

Unit tangent vectors

- 7 For $\vec{r}(t) = \langle t, t^2 \rangle$, find the unit tangent vector $\vec{T}(t)$ at $t = 1$.
 $\vec{v}(t) = \langle 1, 2t \rangle$; at $t = 1$: $\vec{v}(1) = \langle 1, 2 \rangle$, $|\vec{v}(1)| = \sqrt{1 + 4} = \sqrt{5}$. $\vec{T}(1) = \left\langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle$.
- 8 Explain why $\vec{T}(t) = \frac{\vec{v}(t)}{|\vec{v}(t)|}$ always has magnitude 1.
Dividing any nonzero vector by its own magnitude produces a vector with magnitude $\frac{|\vec{v}(t)|}{|\vec{v}(t)|} = 1$ by definition — this is the standard construction of a unit vector in the same direction.

Motion on a circle and centripetal acceleration

- 9 A particle moves along $\vec{r}(t) = \langle 3\cos t, 3\sin t \rangle$. Find $\vec{v}(t)$, $\vec{a}(t)$, and verify that $\vec{a}(t)$ points toward the origin (is a scalar multiple of $-\vec{r}(t)$).
 $\vec{v}(t) = \langle -3\sin t, 3\cos t \rangle$. $\vec{a}(t) = \langle -3\cos t, -3\sin t \rangle = -\vec{r}(t)$. Since $\vec{a}(t)$ is exactly -1 times $\vec{r}(t)$, it points directly toward the origin (centripetal acceleration).
- 10 For the same particle, verify that $\vec{v}(t) \cdot \vec{r}(t) = 0$ for all t , and interpret this geometrically.
 $\vec{v}(t) \cdot \vec{r}(t) = (-3\sin t)(3\cos t) + (3\cos t)(3\sin t) = -9\sin t \cos t + 9\sin t \cos t = 0$. This confirms the velocity vector is always perpendicular to the position vector — consistent with circular motion, where velocity is tangent to the circle.
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Displacement vs. distance for vector motion

- 11 A particle has position $\vec{r}(t) = \langle t^2, t \rangle$ for $t \in [0, 2]$. Find the displacement vector from $t = 0$ to $t = 2$.
 $\vec{r}(0) = \langle 0, 0 \rangle$; $\vec{r}(2) = \langle 4, 2 \rangle$. Displacement $= \vec{r}(2) - \vec{r}(0) = \langle 4, 2 \rangle$.
- 12 Set up (but do not evaluate) the integral for the total distance traveled by the particle $\vec{r}(t) = \langle t^2, t \rangle$ over $t \in [0, 2]$.
 $\vec{v}(t) = \langle 2t, 1 \rangle$. Distance $= \int_0^2 |\vec{v}(t)| dt = \int_0^2 \sqrt{(2t)^2 + 1^2} dt = \int_0^2 \sqrt{4t^2 + 1} dt$.
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Vector operations review

- 13 Given $\vec{u} = \langle 3, -2 \rangle$ and $\vec{w} = \langle -1, 4 \rangle$, find $2\vec{u} - 3\vec{w}$.
 $2\vec{u} = \langle 6, -4 \rangle$; $3\vec{w} = \langle -3, 12 \rangle$. $2\vec{u} - 3\vec{w} = \langle 6 - (-3), -4 - 12 \rangle = \langle 9, -16 \rangle$.
- 14 Find the dot product $\vec{u} \cdot \vec{w}$ for $\vec{u} = \langle 3, -2 \rangle$ and $\vec{w} = \langle -1, 4 \rangle$, and state whether the vectors are perpendicular.
 $\vec{u} \cdot \vec{w} = 3(-1) + (-2)(4) = -3 - 8 = -11$. Since the dot product is not 0, the vectors are NOT perpendicular.
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Curvature-related speed problems

- 15 A particle moves along $\vec{r}(t) = \langle t^3, t^2 \rangle$. Find the times t when the particle's speed is exactly $\sqrt{13}$.
 $\vec{v}(t) = \langle 3t^2, 2t \rangle$. Speed² $= 9t^4 + 4t^2 = 13$. Let $w = t^2$: $9w^2 + 4w - 13 = 0$. Using the quadratic formula:
 $w = \frac{-4 \pm \sqrt{16 + 468}}{18} = \frac{-4 \pm 22}{18}$. Taking the positive root: $w = 1$, so $t^2 = 1 \Rightarrow t = \pm 1$.
- 16 For the particle $\vec{r}(t) = \langle t, \ln t \rangle$ ($t > 0$), find the velocity vector and speed at $t = 1$.
 $\vec{v}(t) = \langle 1, \frac{1}{t} \rangle$. At $t = 1$: $\vec{v}(1) = \langle 1, 1 \rangle$. Speed $= \sqrt{1^2 + 1^2} = \sqrt{2}$.
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Projectile motion as a vector-valued function

- 17 A projectile is launched with position $\vec{r}(t) = \langle 20t, 30t - 4.9t^2 \rangle$ (meters, seconds). Find the velocity vector $\vec{v}(t)$ and the time when the projectile reaches maximum height (where the vertical velocity component is 0).

$\vec{v}(t) = \langle 20, 30 - 9.8t \rangle$. Maximum height occurs when $30 - 9.8t = 0 \Rightarrow t = \frac{30}{9.8} \approx 3.06$ s.

- 18 For the same projectile, find the speed at $t = 0$ (launch) and interpret its two components.

$\vec{v}(0) = \langle 20, 30 \rangle$. Speed $= \sqrt{20^2 + 30^2} = \sqrt{400 + 900} = \sqrt{1300} \approx 36.1$ m/s. The horizontal component (20 m/s) stays constant throughout the flight, while the vertical component (30 m/s initially) decreases due to gravity.

Finding when a particle is at rest

- 19 A particle moves along $\vec{r}(t) = \langle t^3 - 3t, t^2 - 4 \rangle$. Find all times t when the particle is momentarily at rest (both velocity components equal 0 simultaneously).

$\vec{v}(t) = \langle 3t^2 - 3, 2t \rangle$. Need $3t^2 - 3 = 0$ AND $2t = 0$ simultaneously. From the second equation, $t = 0$; check the first: $3(0)^2 - 3 = -3 \neq 0$. There is NO time when both components are zero at once, so the particle is never fully at rest.

- 20 For the same particle, find the times when only the horizontal velocity component is 0.

$3t^2 - 3 = 0 \Rightarrow t^2 = 1 \Rightarrow t = \pm 1$.

Vector-valued functions and arc length

- 21 State the arc length formula for a vector-valued function $\vec{r}(t) = \langle x(t), y(t) \rangle$ on $[a, b]$, and explain how it relates to the speed $|\vec{v}(t)|$.

$L = \int_a^b |\vec{v}(t)| dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$ — arc length is the integral of speed over time, which matches the intuition that distance traveled equals the accumulated speed.

- 22 Find the arc length of $\vec{r}(t) = \langle 4t, 3t \rangle$ for $t \in [0, 5]$ using the formula above.

$\vec{v}(t) = \langle 4, 3 \rangle$, so $|\vec{v}(t)| = \sqrt{16 + 9} = 5$ (constant speed). $L = \int_0^5 5 dt = 25$.